Yuchen Guo

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Education

Northwestern University, USA Sept. 2025 – Dec. 2026 (expected)
Master of Science in Artificial Intelligence, GPA: 4.0/4.0

Hong Kong Baptist University, China Sept. 2021 – Jun. 2025
Bachelor of Science (Honors) in Computer Science and Technology, GPA: 3.5/4.0
First Class Honors, President's Honor Roll

University of Oxford, UK Summer 2023
Summer Visiting in Artificial Intelligence and Machine Learning, GPA: 4.0/4.0
Oxford Outstanding Performance Team

Research Interests

I study intelligence by building multimodal systems that learn and develop through experience and interaction with the world. My current work asks how such systems can expand what they perceive, internalize what they experience, and use their understanding to reason about and act in the world.

Multimodal Perception · Grounded Reasoning · Agent Memory · Continual Learning

Selected Publications

* Equal contribution.

Can Segmentation Models Understand the World? Proactive Affordance Reasoning via Vision-Grounded Chain-of-Thought
Yuchen Guo, Junli Gong, Hongmin Cai, Yiu-ming Cheung, Weifeng Su
EMNLP 2026.

PEAM: Parametric Embodied Agent Memory for Continual Skill Learning via Internalization of Experience in Minecraft
Yuchen Guo, Junli Gong, Weicheng Wang, Hongmin Cai, Yiu-ming Cheung, Weifeng Su
EMNLP 2026 Findings.

SAM3-I: Segment Anything with Instructions
Jingjing Li*, Yue Feng*, **Yuchen Guo***, Jincai Huang*, Wei Ji, Qi Bi, Yongri Piao, Miao Zhang, Xiaoqi Zhao, Qiang Chen, Shihao Zou, Huchuan Lu, Li Cheng
ACL 2026.

Adding Thermal Awareness to Visual Systems in Real-Time via Distilled Diffusion Models
Yuchen Guo, Junli Gong, Wenjun Dong, Hongmin Cai, Yiu-ming Cheung, Weifeng Su
CVPR 2026 Workshop on World Models Meet Active Sensing and Closed-Loop Planning.

Towards Unified Surgical Scene Understanding: Bridging Reasoning and Grounding via MLLMs
Jincai Huang*, Shihao Zou*, **Yuchen Guo**, Jingjing Li, Wei Ji, Kai Wang, Shanshan Wang, Weixin Si
MICCAI 2026 Early Accept.

DAE-Fuse: An Adaptive Discriminative Autoencoder for Multi-Modality Image Fusion
Yuchen Guo*, Ruoxiang Xu*, Rongcheng Li, Weifeng Su
ICME 2025 Oral.

Preprints

Imprint of Experience: How Language Shapes Concept Representations from Pretraining to Continual Learning
Yuchen Guo, Junli Gong, Yiqi Shi, Hongmin Cai, Yiu-ming Cheung, Weifeng Su
Under review at ICLR 2027.

Looking Where Infrared Matters: Instruction-Driven Instance Segmentation in Multispectral Vision
Yuchen Guo, Jingjing Li, Junli Gong, Wei Ji, Shihao Zou, Huidong Xie, Miao Zhang, Yongri Piao, Li Cheng
Under review at ICLR 2027.

Accelerating Reasoning and Grounding for Surgical Intelligence via Multi-Granular Token Compression
Weixin Si, Jincai Huang, Jingjing Li, Wei Ji, **Yuchen Guo**, Kai Wang, Shihao Zou, Yang Gao, Shanshan Wang
Under review at IEEE TMI.

Research & Industry Experience

Yale University & Carnegie Mellon University Sept. 2025 – Present
Research Intern
Advised by Dr. Wei Ji (Yale) and Dr. Jingjing Li (CMU)

- **SAM3-I (ACL 2026):** Co-led the extension of the SAM family from concept-level segmentation to complex instruction following. Developed cascaded adapters, alignment objectives, and a multi-agent data pipeline, improving gIoU by 22.6 points over SAM3 Agent while preserving concept-level capabilities.
- **Looking Where Infrared Matters (ICLR 2027, under review):** Led research moving multimodal perception from passive fusion to active, task-driven evidence acquisition. Proposed instruction-driven RGB-thermal instance segmentation and a framework with learned thermal requests and utility-supervised spatial gating, so models decide when and where thermal evidence helps scene understanding and grounding.

Independent Research Jun. 2025 – Present
Funding and compute support from Northwestern University, Lambda, and Hong Kong Baptist University

- **SegWorld (EMNLP 2026):** Conceived and led research on whether segmentation models understand the world or merely label it. Proposed a framework that proactively observes the scene, then reasons via vision-grounded chain-of-thought from human-centric intent to affordance-bearing part (+13.2 mIoU on our Intent2Part).
- **PEAM (EMNLP 2026 Findings):** Developed an embodied agent memory framework in Minecraft that turns experience from inference-time retrieval into parameter-resident skills, using Mixture-of-Experts LoRA with isolated adapters and self-triggered consolidation for continual learning without forgetting.

The Vanguard Group, Inc. Sept. 2026 – Present
Graduate Capstone Researcher
Advised by Dr. Sagarika Suresh, Leizer Teran, Dr. Sam McDermott and Dr. James Belasco

- **Multimodal Financial Agent and Forward-Deployed Frontier LLMs:** Developing an agent that reasons jointly over charts, tables, and narrative text in financial documents, grounding its claims in source material and flagging chart structures prone to misinterpretation. Benchmarking open-weight (Gemma) and proprietary (Claude, GPT, Gemini) models on interpretation accuracy, grounding quality, and hallucination risk.

SIAT MMLab, Chinese Academy of Sciences Jan. 2024 – Jan. 2025; Sept. 2025 – Sept. 2026
Research Intern, Member of XPixel Group
Advised by Prof. Chao Dong and Dr. Shihao Zou

- **SurgMLLM & EffSurgMLLM (MICCAI 2026; IEEE TMI, under review):** Contributed to advancing surgical intelligence from isolated recognition tasks to unified, efficient reasoning and pixel-level grounding via MLLM. Developed methods integrating surgical phase recognition, instrument–verb–target reasoning, and entity grounding, with spatial–temporal–semantic token compression reducing visual tokens by 84.4% while maintaining comparable task performance.
- **Video Generation via Human Image Animation:** Extended MoMask with audio conditioning to generate motion guidance (depth, normal, semantic maps, and skeletons) from audio and text, which drives a diffusion + ControlNet framework to animate a reference image into temporally consistent video.

Institute for AI Industry Research (AIR), Tsinghua University Dec. 2024 – Mar. 2025
Research Intern
Advised by Prof. Yilun Chen and Dr. Pengfei Li

- **World Model for Long-Tail Autonomous Driving:** Developed a driving world model targeting long-tail scenarios in end-to-end autonomous driving, such as distinguishing stalled vehicles from those temporarily stopped in traffic. Modeled future scene evolution from multimodal inputs, enabling the planner to anticipate whether surrounding agents will move and decide between waiting and bypassing.

Honors & Service

MSAI Conference Funding, Northwestern University 2026
Medical Valley Scholarship, China 2024
Reviewer: AAAI 2027, ICLR 2027, ACL 2026, EMNLP 2026, CVPR 2026, ICME 2025
Teaching Assistant: COMP4223 Deep Learning for Computer Vision, 2024-2025